



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Fakulti
Komputeran**

Course / Subject	SCSE1203 Software Engineering Principle
Section	04
Team Name	Team 6
Product / Project Title	NexScholar System
Sprint Duration	Week 13 / Sprint #1 (4 days)

1. Sprint #1 Overview

During Sprint #1 (Week 13), Team 6 initiated the first development cycle of the NexScholar System using the Scrum framework and Taiga project management tool. The sprint focused on establishing the core functionality of the system, including AI-based research matching, research discovery, personalized onboarding, and interactive tutorial features. The team collaborated closely with the Product Owner to define and prioritize four main epics and their associated user stories.

A complete Product Backlog consisting of eleven user stories was created and organized under four epics: E1 (AI Matching & Search), E2 (Research Discovery & Collaboration), E3 (Interactive Tutorial System), and E4 (Personalized Onboarding). Based on business value, dependencies, and sprint capacity, four user stories (US101, US201, US301, and US401) were selected for Sprint #1, representing a total of 26 story points.

User stories and tasks were distributed among team members according to their assigned roles and responsibilities. Throughout the four-day sprint, the team continuously updated the Product Backlog, Sprint Backlog, and Sprint Taskboard in Taiga to monitor progress. Several design-related tasks were completed, implementation activities were actively in progress, and testing tasks were prepared for subsequent validation stages. Sprint progress was monitored through the Sprint Burndown Chart, while Daily Stand-up Meetings were conducted to discuss progress, identify blockers, and ensure alignment among team members. Evidence of Scrum activities and meetings was documented throughout the sprint [1].

2. Sprint #1 Description

2.1 Prioritize Epics with Product Owner

The team conducted a prioritization meeting with the Product Owner via Google Meet to identify and organize the key functional requirements of the NexScholar System. During the discussion, the Product Owner reviewed the project vision and helped prioritize the system features according to their business value and importance to end users.

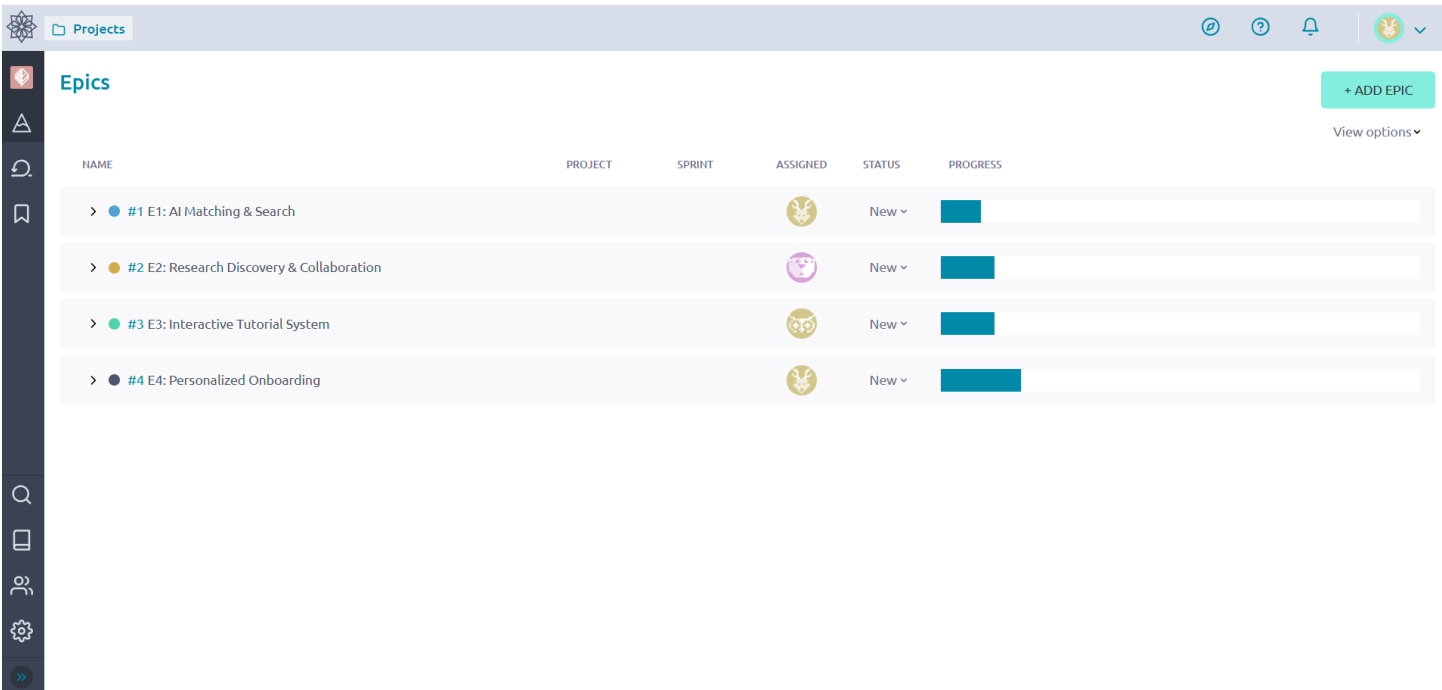
Based on the prioritization session, four major epics were identified: E1 (AI Matching & Search), E2 (Research Discovery & Collaboration), E3 (Interactive Tutorial System), and E4 (Personalized Onboarding). These epics represent the core functional areas of the system and were further decomposed into eleven user stories that formed the Product Backlog.

Considering the sprint duration and team capacity, four user stories were selected for Sprint #1: US101 (Research Preferences During Registration), US201 (Student Evaluation and Discovery), US301 (First-Time User Tutorial Popup), and US401 (Personalized Onboarding Questions), representing a total of 26 story points.

These user stories were selected because they establish the foundation of the user experience by supporting user onboarding, research discovery, and platform guidance. In addition, they provide the necessary groundwork for future AI-based recommendation, collaboration, and personalization features. The selected user stories were added to Sprint 1 and managed using Taiga for sprint planning, task tracking, and progress monitoring.

Figure 2.1 shows the four prioritized epics identified during the Product Owner meeting. The epics represent the major functional areas of the NexScholar System: E1 (AI Matching & Search), E2 (Research Discovery & Collaboration), E3 (Interactive Tutorial System), and E4 (Personalized Onboarding). These epics were established to organize the system requirements into manageable functional groups and guide sprint planning activities.

Figure 2.1.1: Prioritized Epics for the NexScholar System in Taiga



2.2 Distribute User Stories Among Team Members

Tasks were distributed across the team's four members according to specialization:

- **Leen Milhem (Product Owner)** was assigned to **US101 – Research Preferences During Registration**. Responsibilities included designing the onboarding form, updating the registration database schema, implementing preference validation, and conducting testing activities.
- **Mohammad Sameera (Scrum Master)** was assigned to **US201 – Student Evaluation and Discovery**. Responsibilities included designing the profile preview interface, retrieving profile data, and validating the student discovery workflow.
- **Shadin Hanafi (Frontend Developer)** was assigned to **US301 – First-Time User Tutorial Popup**. Responsibilities included designing the onboarding workflow, creating tutorial popups, and implementing tutorial navigation features.
- **Nour Gadallah (UX Developer)** was assigned to **US401 – Personalized Onboarding Questions**. Responsibilities included creating the preference questionnaire, storing user preferences, and reviewing the onboarding experience.

The team decided to assign one primary user story from each epic to a specific team member rather than distributing tasks across multiple epics. This approach ensured clear ownership, accountability, and consistency throughout Sprint #1. Since each selected user story represented a different functional area of the NexScholar System, assigning a dedicated team member to each story allowed for more focused development, easier progress tracking, and efficient task management. Collaboration was still maintained through Daily Stand-up Meetings, sprint discussions, and task reviews, enabling team members to provide feedback and support when required.

The assigned tasks were created and managed in Taiga, allowing the team to track progress, update task status, and monitor sprint activities throughout Sprint #1.

Figure 2.2.1 US101 assigned to Leen Milhem with associated Sprint #1 tasks.

The screenshot shows a Taiga user story card for '#6 US101 – Research Preferences During Registration'. The card is assigned to Leen Milhem and includes a list of tasks. The tasks are: '#49 Design onboarding form' (Closed), '#20 Update database schema' (In progress), '#21 Implement validation' (In progress), and '#22 Testing' (New). The card also shows a points breakdown: UX (5), Design (?), Front (?), Back (?), and a total of 5 points.

Task	Status	Assignee
#49 Design onboarding form	Closed	Leen Milhem
#20 Update database schema	In progress	Leen Milhem
#21 Implement validation	In progress	Leen Milhem
#22 Testing	New	Leen Milhem

Figure 2.2.2. US201 assigned to Mohammad Sameera with associated Sprint #1 tasks

#10 US201 – Student Evaluation and Discovery
USER STORY

This user story belongs to **#2 E2: Research Discovery & Collaboration** 100%

[Link to epic](#)

[TASKBOARD](#)

[Add tag +](#)

Created by Leen Milhem 12 Jun 2026 13:46

As a lecturer, I want to evaluate and discover students based on their profiles and interests so that I can identify suitable research collaborators.

0 Attachments

Drop attachments here!

Tasks

Task ID	Task Description	Status	Assignee
#23	Design popup-UI	Closed	Mohamm...
#24	Retrieve profile data	In progress	Mohamm...
#25	Testing	New	Mohamm...

POINTS

Category	Points
UX	?
Design	?
Front	5
Back	?
total points	5

ASSIGNED

Mohammad Sumera

+ Add assigned Assign to me

WATCHERS

+ Add watchers Watch

Figure 2.2.3 US301 assigned to Shadin Hanafi with associated Sprint #1 tasks.

#13 US301 – First-Time User Tutorial Popup
USER STORY

This user story belongs to **#3 E3: Interactive Tutorial System** 100%

[Link to epic](#)

[TASKBOARD](#)

[Add tag +](#)

Created by Leen Milhem 12 Jun 2026 13:49

As a first-time user, I want a tutorial popup when I enter the platform so that I can understand the main features and navigation.

0 Attachments

Drop attachments here!

Tasks

Task ID	Task Description	Status	Assignee
#26	Design onboarding workflow	Closed	Shadin H...
#27	Create tutorial popups	In progress	Shadin H...
#28	Testing	New	Shadin H...

POINTS

Category	Points
UX	?
Design	?
Front	?
Back	8
total points	8

ASSIGNED

Shadin Hanafi

+ Add assigned Assign to me

WATCHERS

+ Add watchers Watch

Figure 2.2.4 US401 assigned to Nour Gadallah with associated Sprint #1 tasks.

#16 US401 – Personalized Onboarding Questions
USER STORY

This user story belongs to **#4 E4: Personalized Onboarding** 100%

[Link to epic](#)

[TASKBOARD](#)

[Add tag +](#)

Created by Leen Milhem 12 Jun 2026 13:52

As a user, I want personalized onboarding questions so that the platform can better understand my interests and provide relevant recommendations.

0 Attachments

Drop attachments here!

Tasks

Task ID	Task Description	Status	Assignee
#29	Create preference-questionnaire	Closed	nour gad...
#30	Store user preferences	In progress	nour gad...
#31	Testing	New	nour gad...

POINTS

Category	Points
UX	?
Design	?
Front	8
Back	?
total points	8

ASSIGNED

nour gadallah mohammedali

+ Add assigned Assign to me

WATCHERS

+ Add watchers Watch

2.3 Conduct Daily Stand-up Meetings

The team conducted a Daily Stand-up Meeting during Sprint #1 to review progress, discuss ongoing tasks, and identify any blockers. During the meeting, each team member shared what had been completed, what would be worked on next, and any challenges encountered. The discussion was based on the Sprint Taskboard in Taiga to ensure transparency and alignment among team members. Daily stand-up meetings are an important Scrum practice that helps improve communication, transparency, and team coordination [1].

Table 3.1 Stand up meeting

Date / Time	6 June 2026, 9:00 PM
Team Members Absent during stand up meeting	None
Main Progress Discussed	Completion of onboarding questionnaire design, progress on student discovery workflows, tutorial popup development, onboarding flow creation, and updates to task statuses in Taiga.
Meeting Recording Link	https://youtube.com/watch?v=Ja568l8CTmc&si=talRVHHoygMgFw4y

2.4 Update Product Backlog Item (PBI) and Sprint Backlog

The Product Backlog was created based on the prioritized epics and consisted of eleven user stories representing the functional requirements of the NexScholar System. The user stories were organized under four epics: E1 (AI Matching & Search), E2 (Research Discovery & Collaboration), E3 (Interactive Tutorial System), and E4 (Personalized Onboarding). The Product Backlog served as the primary source of requirements and was continuously refined throughout the sprint.

Based on the Product Owner's prioritization and the team's sprint capacity, four user stories were selected for Sprint #1: US101 (Research Preferences During Registration), US201 (Student Evaluation and Discovery), US301 (First-Time User Tutorial Popup), and US401 (Personalized Onboarding Questions). These user stories represented a total of 26 story points and formed the Sprint Backlog.

The Product Backlog and Sprint Backlog were maintained in Taiga to ensure transparency and effective sprint planning. The backlog structure allowed the team to track requirements, prioritize work items, and manage sprint activities throughout Sprint #1.

Figure 2.4.1. Product Backlog showing all epics and associated user stories of the NexScholar System.

NAME	PROJECT	SPRINT	ASSIGNED	STATUS	PROGRESS
#1 E1: AI Matching & Search				New	
#6 US101 – Research Preferences During Registration		Sprint 1		New	
#8 US102 – Profile-Based Quick Search Suggestions				New	
#9 US103 – AI / Keyword Search Toggle				New	
#2 E2: Research Discovery & Collaboration				New	
#10 US201 – Student Evaluation and Discovery		Sprint 1		New	
#11 US202 – Profile Preview Popup				New	
#12 US203 – External Communication Links				New	
#3 E3: Interactive Tutorial System				New	
#13 US301 – First-Time User Tutorial Popup		Sprint 1		New	
#14 US302 – New Feature Tutorial Popup				New	
#15 US303 – Video and Visual Tutorials				New	
#4 E4: Personalized Onboarding				New	
#16 US401 – Personalized Onboarding Questions		Sprint 1		New	
#17 US402 – Tutorial Recommendations				New	

Figure 2.4.2. Sprint Backlog containing the selected user stories for Sprint #1.

1 SPRINTS	Add +
Sprint 1 04 Jun 2026-07 Jun 2026 0 closed 26 total	
#16 US401 – Personalized Onboarding Questions ●	8
#13 US301 – First-Time User Tutorial Popup ●	8
#10 US201 – Student Evaluation and Discovery ●	5
#6 US101 – Research Preferences During Registration ●	5
SPRINT TASKBOARD	

2.5 Update Burndown Chart

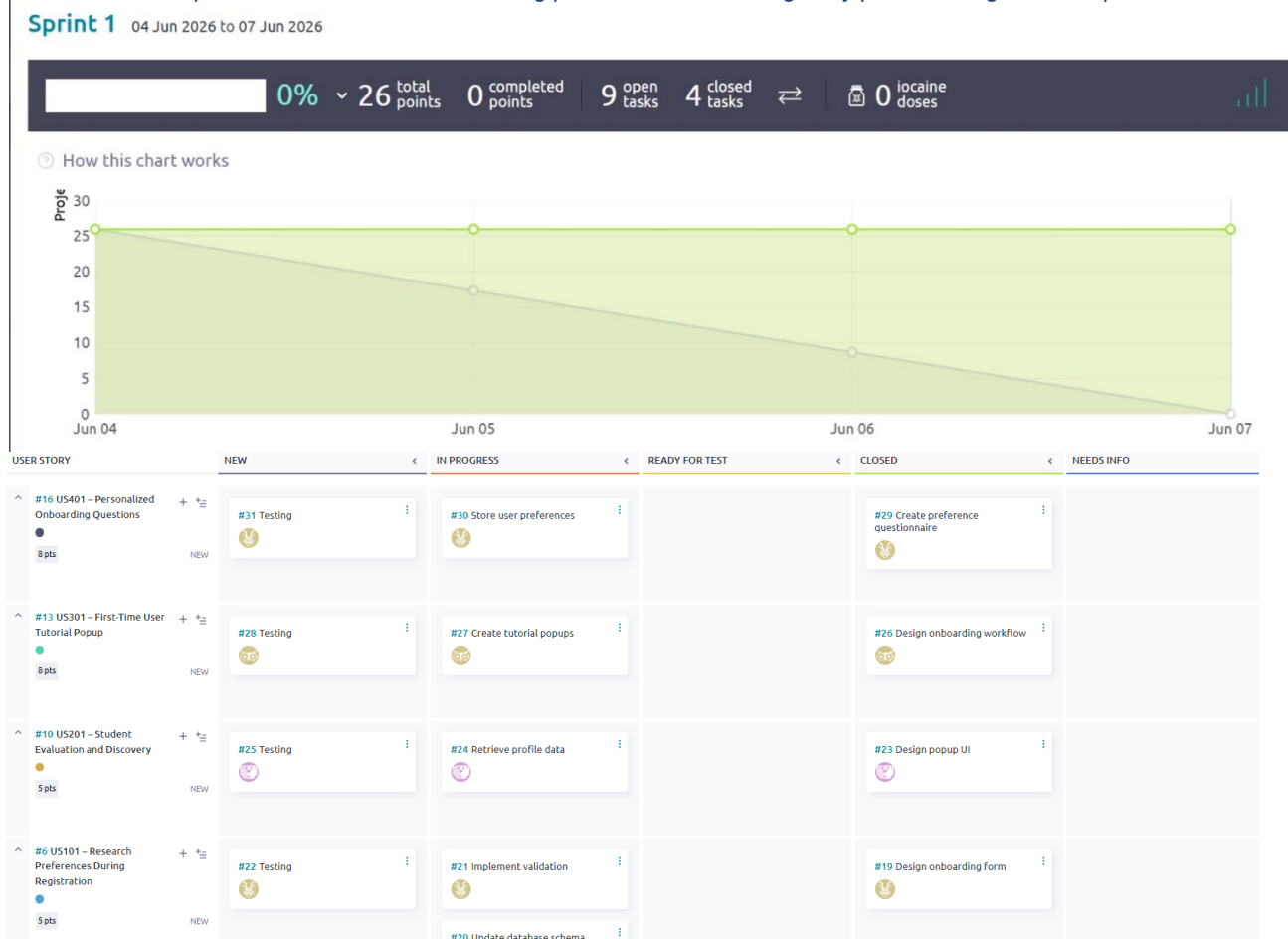
The Sprint #1 burndown chart was used to monitor the team's progress and remaining workload throughout the sprint. The selected user stories for Sprint #1 represented a total of 26 story points, consisting of US101 (5 points), US201 (5 points), US301 (8 points), and US401 (8 points).

Figure 2.8 illustrates the Sprint #1 burndown chart generated in Taiga. The chart provided visibility into the planned workload and remaining effort during the four-day sprint. Team members used the chart to monitor sprint progress and ensure that tasks were progressing according to the sprint plan.

Although several tasks were completed and others were actively in progress, the selected user stories had not yet been fully completed at the time of monitoring. As a result, the remaining story points shown in the burndown chart remained unchanged throughout the sprint. Nevertheless, the chart served as an effective Scrum monitoring tool by enabling the team to compare planned work against actual progress and identify areas requiring additional attention.

The burndown chart also supported transparency and continuous tracking of sprint activities, helping the team assess progress toward achieving the Sprint #1 objectives. The Sprint Taskboard enabled the team to monitor work progress, identify pending activities, and ensure that tasks moved systematically through the workflow stages of New, In Progress, Ready for Test, and Closed [2].

2.5.1 Sprint #1 Burndown Chart showing planned and remaining story points throughout the sprint.



2.6 Conduct Scrum Review for Sprint #1

As stakeholder feedback isn't officially collected at this introductory milestone, the team conducted an internal review session to demo the UX flows and architectural progress.

Table 3.2 Sprint Review Summary

Review Date	June 7, 2026
Completed User Stories	No user stories were fully completed during Sprint #1 . However, several design-related tasks under US101 , US201 , US301 , and US401 were successfully completed.
Work Demonstrated	Demonstration of the onboarding form design, preference questionnaire, student profile interface design, tutorial workflow design, and updates to the registration database structure.
Unfinished Work	US301 : Tutorial popup implementation remained in progress and testing activities had not yet started. US401 : User preference storage functionality was still under development, and testing tasks remained pending at the end of Sprint #1.
Issues Encountered	The four-day sprint duration limited development and testing activities. Several implementation and validation tasks required additional time, resulting in pending testing activities.
Internal Feedback	Team members agreed that task ownership and progress tracking through Taiga improved visibility and coordination. The team recommended allocating additional time for testing and validation activities in future sprints.
Next Action	Carry forward incomplete tasks to Sprint #2, complete testing activities, finalize remaining implementation work, and continue refining onboarding and tutorial features.

2.7 Conduct Scrum Retrospective

The retrospective was completed on June 7, 2026, to evaluate internal team dynamics and tools.

Table 3.3 Sprint Retrospective Summary

Retrospective Date	June 7, 2026.
What Went Well?	High collaboration on design tasks; Taiga board was updated uniformly; great cross-functional support between frontend and backend pairs.
What Problems Occurred?	Task sizing was a bit too large for a 4-day sprint; initial setup delays with the local environment.
What Should Be Improved?	Break tasks into even smaller sub-tasks (less than 3 story points each) to make the burn-down smoother; start development tasks faster.
Improvements for next sprint	Action Item: Pre-configure the joint repository environments before Day 1 of Sprint #2

3. Conclusion

The biggest collective challenge during Sprint #1 was adjusting to the rapid rhythm of the Scrum framework within a highly compressed 4-day timeline. Coordinating synchronous stand-up sessions alongside demanding university schedules required deliberate scheduling effort.

However, stabilizing our workflow tracking on Taiga during this initial milestone sets a solid baseline for velocity optimization in subsequent sprints.

References

APA style references

[1] Schwaber, K., & Beedle, M. (2002). *Agile software development with Scrum*. Prentice Hall.

[2] Taiga Agile LLC. (n.d.). *Taiga: Agile project management platform*. <https://www.taiga.io/>.